

Figure 15.3 Rotary Mixer—Pulvimixer Type

the forward speed of the unit. In the second method, an asphalt distributor sprays the asphalt on the aggregate ahead of the mobile mixer; the materials are drawn into the mixing chamber as the machine moves forward.

When this type of mixing is used, the following steps are recommended:

1. Spread the aggregate to a uniform grade and cross section with a motor grader.
2. Thoroughly mix the aggregate with one or more passes of the mixer. (At the time the asphalt is applied, the moisture content of the aggregate should not exceed 3 percent for cutback asphalt, unless laboratory tests indicate that a higher moisture content will not be harmful when the asphalt is added.) (For emulsion mixes, see "Methods of Preparing Mixes" in this chapter.)
3. Add asphalt in increments of about 0.50 gallon per square yard (2.25 liters per square meter) until the total required amount of asphalt is applied and mixed into the aggregates. A total of 0.4 to 0.6 gallon per square yard per inch (0.7–1.1 liter per square meter per 10 millimeters) of the compacted thickness is usually necessary.
4. Make one or more passes of the mixer between applications of asphalt, as necessary to ensure thorough mixing.
5. Maintain the surface true to grade and cross section by using a motor grader during the mixing operations.
6. Aerate the mixture by additional manipulation, if necessary.

Travel-Plant Mixing

Travel plants are self-propelled pugmill plants that mix the aggregates with asphalt, which is applied at a controlled rate, as they move along the

road. One type of travel plant has an aggregate receiving hopper, an asphalt storage tank, pugmill mixer, and floating screed. Recycled material, virgin aggregate, or a blend of the two can be fed (by truck or low-lift loader from the windrow) into the plant for mixing with asphalt (Figure 15.4). The completed mixture is discharged at the rear, and the thickness can be adjusted to any desired depth.

Travel plants should be capable of proportioning and thoroughly mixing asphalt and aggregates, uniformly dispersing the asphalt, and adequately coating the aggregate particles.

The asphalt application rate for the windrow pickup plant must be accurately matched with the width and thickness of the course, and the forward speed of the mixer. If all the asphalt cannot be incorporated in one pass, the windrow should be split into two or more windrows and the asphalt added to each windrow for mixing. Depending on the type of travel plant and mix components, further mixing of the recycled material may be necessary with a motor grader. This ensures that all the windrowed material is incorporated into the mix. It also aerates the mixture. The number of passes with the motor grader varies with job conditions.

Travel-plant mixing offers the advantage of achieving closer control of the mixing operation than is possible with blade mixing. The proportioning of the asphalt with the aggregate, as well as the uniformity with which the two can be incorporated, is of great importance.

The mixing machine moves along the windrow, adding the asphalt and mixing it into the windrow as it moves forward. The rate of flow per minute at which the asphalt pump is adjusted may be determined by the following:

$$E = AS$$

where

E = discharge rate of asphalt, gallons per minute (liters per minute)
A = amount of asphalt, gallons per linear foot (liters per linear meter)
S = forward travel speed, feet per minute (meters per minute)

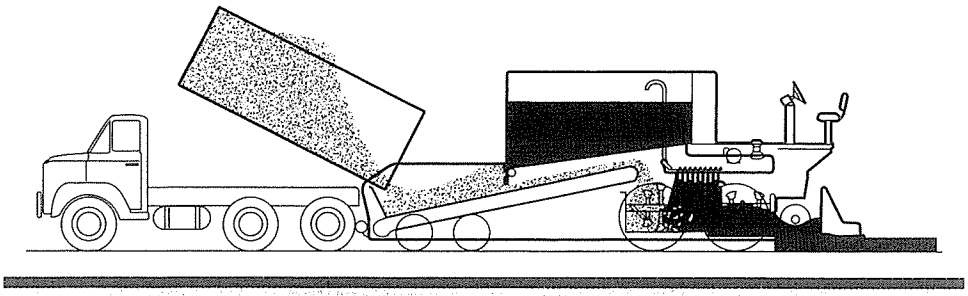


Figure 15.4 Schematic of Travel Plant

Example

A windrow of aggregate is to have 5.3 gallons (20 liters) of MC-250 asphalt per 3.28 per linear foot (linear meter) mixed into it with a travel-plant mixer. For a forward speed of 16.4 feet per minute (5 meters per minute) the pumping rate for the asphalt is found by using the above formula.

Customary Units	Metric Units
$E = \frac{5.3}{3.28}(1.64) = 26.5$ gal/min	$E = 20\ (5) = 100$L/min

If the windrow is too large for the travel plant, it may be split into two or more windrows and asphalt added and mixed to each separately. Further mixing of the windrow is often necessary and usually accomplished with the motor grader. This process also affords extra aeration of the mix.

Stationary Mixing Facilities

Mixes using emulsified and cutback asphalt prepared at a stationary mixing facility offer certain advantages over road-mixing methods. Weather conditions are not as much a factor in a stationary mixing facility. Mixing that would otherwise require hours of windrow manipulation is accomplished in a few minutes.

The stationary mixing facility provides good control over both the proportions of materials in the mix and the mixing time. Uniformity of mixes is practically assured. Processing in a central plant allows a more viscous asphalt to be used. Often, enough aeration has been accomplished in the mixing operation—dumping and spreading mix in the roadway—to permit immediate compaction with rolling equipment.

Often, these advantages make facility-mixing methods more economical for producing and placing mixes in the roadway than road-mixing methods.

Cold mix facility setups may vary depending on the quality and the type of mix being produced. At the very least, the setup should consist of a mixer, an asphalt storage tank, an asphalt metering pump, piping and spray bar equipment, a spray bar for feeding water and additives, controls for adjusting and monitoring the various components, a conveyor for feeding the aggregate, and a power source. It may also include one or more aggregate bins, proportioning aggregate feeders, a scalping screen, an aggregate load-sensing device, and a surge bin or storage silo. Batch-type pugmills can be used. However, this mix production is ideally suited for continuous mixers, which are used almost exclusively.

The production of high quality cold mixes for heavy duty pavements requires a well controlled facility setup to ensure success. In addition to carefully monitored and controlled blending of aggregate, asphalt, and in some cases water, the mixer should be a type that permits variation in mixing time. Surge hoppers or storage silos are highly desirable because they minimize facility shutdowns and improve mix uniformity. Good stockpiling and loader techniques can also be used in lieu of surge bins or silos. A typical cold mix facility is shown in Figure 15.5. The steps in the cold mix design procedure are shown in Figure 15.6.



Figure 15.5 Cold Mix Continuous Mixing Facility

Aeration/Spreading/Compaction

Aeration of open-graded aggregate mixtures is usually not critical. However, as the presence of fine aggregate increases and the mix becomes dense-graded, ample aeration is necessary. Proper aeration for cutback asphalt is usually reached when the volatiles have been reduced to about 50 percent of that contained in the original asphalt and the moisture content does not exceed 2 percent by weight of the total mixture. For emulsified asphalt mixtures, the moisture content should be reduced to not more than 2–5 percent by weight of the total mix. In the case of sands and sandy soils used as a base course and surfaced within 30 days, volatiles in the mix should be reduced by at least two-thirds.

The point at which adequate aeration is achieved is often based on experience and judgment. Experience with the particular mixture being prepared and trial rolling have practical value for determining the presence of moisture or volatiles when test data are not immediately available. (AASHTO T 110) *Standard Method of Test for Moisture or Volatile Distillates in Bituminous Mixtures* can also be used. This test uses a metal still (similar to the one used for the distillation of emulsified asphalt), condenser, and a glass trap. A light solvent is added to a sample of the uncompacted mix in the still and then heated.

An experienced motor-grader operator will watch as the material rolls off the end of the plow board. If the material displays sheen, the mix has not had adequate aeration. Once the material displays a dull or flat color,

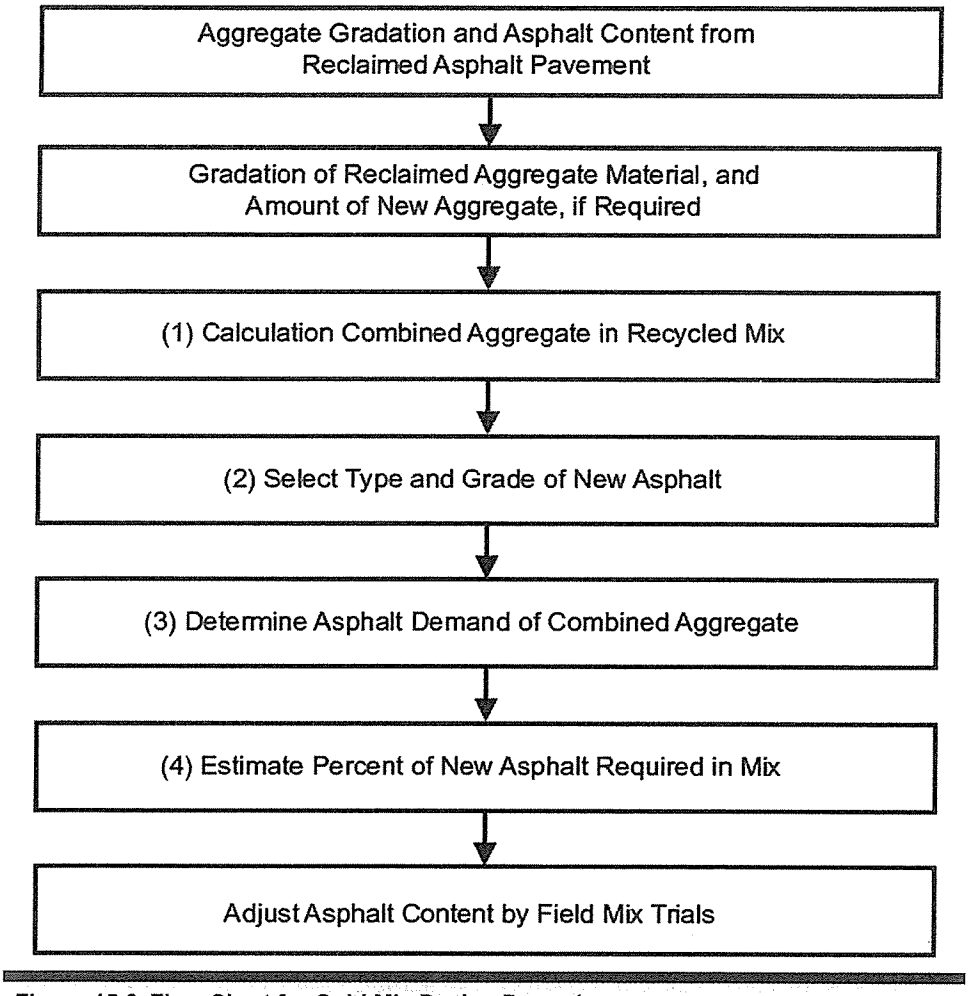


Figure 15.6 Flow Chart for Cold Mix Design Procedure

it is properly aerated. A moisture-content check as described above will confirm the mixture is ready for placement and compaction.

Mixtures that have been windrowed by a travel plant or by blade mixing are spread and shaped to the desired cross section by successive passes with a motor grader. Mixtures blended by rotary cross-shaft mixers, although in place on the roadway, generally require additional shaping. This is also done with the motor grader.

Mixtures that do not require aeration may be spread to the required thickness immediately after mixing. Mixtures that require aeration are generally deposited upon the roadbed in a windrow. Because of a tendency to leave a hump in the road when blade spreading from a centerline windrow, it is better to move the windrow to the side before spreading. Blade spreading should be done in successive layers, with no layer thinner than three times the maximum particle size.

Construction should not continue during rainfall, nor should it begin when rain is expected. Also, the emulsified-asphalt base should not be placed if

the ambient temperature is below 50°F (10°C). Emulsified-asphalt mixes gain stability as the fluids (mainly water) evaporate. It is important not to retard this process. Therefore, lift thickness must be limited by the rate at which the mixture loses its fluids. The type of emulsified asphalt, the water content of the mix, the gradation and temperature of the aggregate, wind velocity, ambient temperature, and humidity all affect evaporation.

Although each job has its own combination of these factors, experience has shown that under the best conditions, a dense-graded mix should be placed in compacted thicknesses no greater than 3 inches (75 millimeters). Open-graded mix may be placed in layers up to 6 inches (150 millimeters). When multiple lifts are required, some curing time must be allowed between successive lifts. Curing time varies with the rate of evaporation. However, a second lift can usually be placed after two to five days under good curing conditions.

A delay in initial rolling may be necessary, especially with dense-graded mixes. Rolling seals the pavement as it reduces the voids in the mix. Premature rolling retards evaporation of the water in the mix and greatly extends the time required for the mix to reach required density. The mix must be allowed to develop sufficient cohesion to support the rollers. If rolling is delayed too long, however, compaction will be difficult. In some cases, the developing asphalt-aggregate bond will be irreversibly broken; therefore, timing is important. Rollers may be selected as indicated in Table 15.6.

Because the tires of a motor grader compact the freshly spread mix, the tracks will appear as ruts in the finished mat unless there is adequate rolling between each successive lift. Therefore, the roller should follow directly behind the motor grader in order to eliminate wheel tracks.

If the asphalt mixture exhibits undue rutting or shoving at any time during compaction, rolling should be stopped. Compaction should not be attempted until there is a reduction in moisture content, either naturally or by mechanical aeration.

After one lift is thoroughly compacted and cured, another lift may be placed on it. The operation should be repeated as necessary to bring the road to required grade and cross section. For a smooth riding surface, the motor grader should be used to trim and level as the rollers complete compaction of the lift.

After the mat is shaped to the final cross section, it is rolled with a steel-wheel roller until the roller marks are eliminated.

Table 15.6
Types of Rollers Suitable for the Compaction of Cold Mix Asphalt Bases

Type of Cold Mix Asphalt Base	Stage of Compaction		
	Breakdown Rolling	Intermediate Rolling	Finish Rolling
Dense-Graded	Steel-wheel Pneumatic Vibratory	Steel-wheel Pneumatic Vibratory	Steel-wheel
Open-Graded	Steel-wheel*	Steel-wheel Pneumatic** Vibratory	Steel-wheel

* Breakdown rolling of open-graded emulsified-asphalt base is often facilitated by adding a small amount of detergent to the sprinkler system water to prevent pickup

** If intermediate rolling of an open-graded emulsified-asphalt base is to be with a pneumatic-tire roller, or if traffic must use a recently placed mix, a chokestone application of from 4–6 pounds per yard (2.2–3.3 kg/m^2) of suitable aggregate should be applied. Stone screenings from a crushing operation or concrete sand are among those suitable.

15.2 Hydraulic Structures

Asphalt has been used in hydraulic applications for several thousand years. Archaeological excavations show that asphalt was used as waterproofing material for temple baths, aqueducts, and water tanks as long as 5,000 years ago.

The use of asphalt for the control, conservation, and storage of water has expanded in past decades.

Areas of growing importance include the following:

- Water conservation, to meet the increasing demand for water supplies
- Irrigation, to open up new lands for cultivation
- Flood control, to protect lives and investments in personal property and industry
- Beach erosion, to preserve beaches and protect waterfront property

15.2.1 Application/Uses for Hydraulic Structures

The following list is a compilation of some applications and uses of asphalt in hydraulic structures.

- *Breakwater*—a structure protecting a shore area, harbor, anchorage, or basin from waves
- *Canal*—a watercourse cut through a land area and used for either irrigation or navigation
- *Catch basin, pond, or reservoir*—a large, shallow holding area designed to collect rainwater. It usually discharges the collected water into a storage tank (see Figure 15.7)
- *Dam*—an artificial barrier of earth, rock, masonry, or other material across a channel to impound water or to obstruct its flow
- *Dike*—a wall or mound built around an area to impound water or to prevent flooding. If used to protect from flooding, it is usually called a levee
- *Embankment*—an artificial bank, mound, or dike built for a specific purpose, such as to hold backwater, or to carry a roadway
- *Groin*—a shore-protective structure (such as a pier or structure of stone, rubble) extending out from and perpendicular to the shoreline to trap lateral drift sand and retard shore erosion, or to protect a harbor or to influence water currents or tides
- *Levee*—a dike or embankment to protect a land area from flooding
- *Sanitary landfill*—an area of land designed for the disposal of refuse in accordance with sound engineering methods whereby the refuse

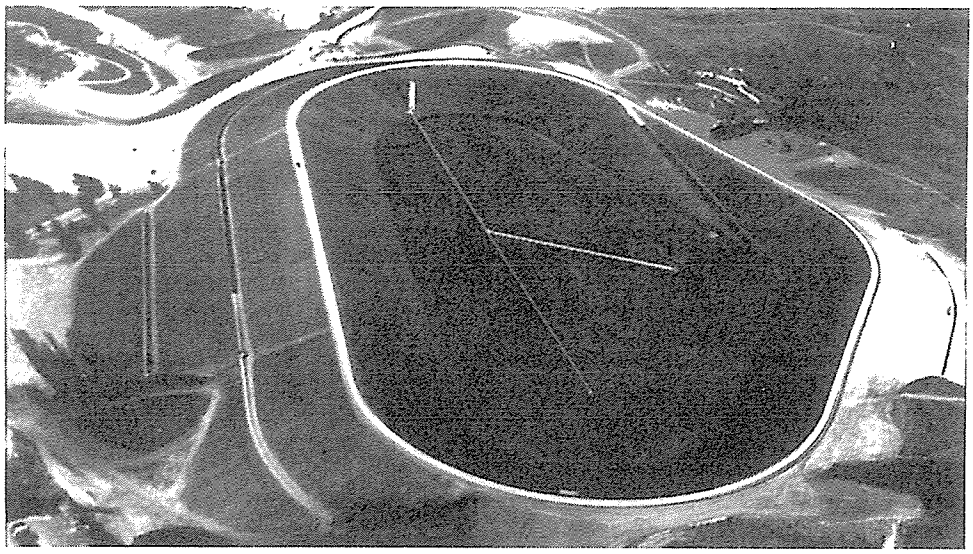


Figure 15.7 Asphalt-Lined Reservoir

is confined to the smallest practical area, reduced to the smallest possible volume, and covered with earth periodically

- *Seawall*—a structure separating land and water areas, primarily designed to prevent erosion and other damage by wave action or inundation
- *Sewage lagoon*—a large, shallow reservoir or pond for the treatment of sewage or industrial waste. Natural biological (aerobic) processes that require oxygen and sunlight accomplish treatment.

Asphalt Materials and their Functions

Many types of asphalt materials are used in hydraulic applications. Each type can be classified in one of the following distinct categories:

- Impermeable asphalt mixes
- Porous asphalt mixes
- Asphalt mastics
- Asphalt cement

These materials can be used in various forms to perform one or more of the following functions:

- To waterproof a structure
- To protect a structure
- To reinforce a structure

Table 15.7 shows how each type of asphalt material may be used to perform these functions.

Impermeable Mixes

The most common way of using asphalt to make a hydraulic structure watertight is to pave the surface with hot mix asphalt (HMA), which forms an impermeable lining when compacted. HMA mixes used for this purpose

Table 15.7
Application Forms of Asphaltic Materials for Hydraulic Structures

	Function		
Material	**Waterproof**	**Protect**	**Reinforce**
Impermeable Asphalt Mixes	Surface or exposed linings	Surface or exposed linings	
Porous Asphalt Mixes		Permeable surface linings. Open-graded subsurface drainage layers	
Asphalt Membranes	Surface linings Seal coats Cutoff walls Dam cores Subsurface injection	Surface linings Seal coats	Grouting Penetration treatments

are generally similar to those used for roadway paving but with a higher dust and asphalt content, and are prepared in an asphalt mixing plant. Standard roadway construction equipment is usually modified for this type of paving. In some instances, special paving equipment has been devised. Linings of impermeable asphalt mixes consisting of well-graded aggregates provide an erosion-resistant, durable surface. Even when the waterproofing of an embankment is not required, impermeable asphalt revetments or facings resist scouring, weed growth, and other damage.

Porous Asphalt Mixes

There are two types of porous mixes: permeable and open-graded. Permeable asphalt mixes are used to line reservoirs that are made watertight by compacted layers of earth or clay. The asphalt linings protect the earth embankment from erosion and at the same time permit the free flow of water through the lining, thereby eliminating the danger of hydrostatic back pressures developing under the lining. Open-graded mixes are used to provide drainage layers beneath watertight linings. Drainage layers are designed to dispose of seepage or leakage that may occur in the watertight lining. They are also used to drain away groundwater seepage that would otherwise build up hydrostatic pressures underneath the watertight lining. Open-graded mixes, in addition to providing a subsurface drainage layer, add to the stability and overall strength of the lining.

Asphalt Seals

An asphalt membrane lining consists of a continuous layer of asphalt cement sprayed over the surface being waterproofed. It is usually covered with a layer of soil, gravel, or other material to hold the lining in place, to prevent weathering (oxidation) of the asphalt, and to protect it from mechanical damage. The asphalt is usually sprayed in two applications at a total rate of about 1½ gallons per square yard (6.8 liter per square meter) to

obtain a layer thickness of about ¼ inch (6 millimeters). Special air-blown asphalts have been developed for this type of application. These asphalts have a high softening point of 175°–200°F (79°–93°C) and a ductility as low as 3.5 centimeters.

The subgrade should first be sterilized to deter weed growth. The surface should be free of lumps, clods, and large rocks. It is rolled before asphalt is applied. Application temperatures should be from 350°F to 425°F (177°C–218°C). Cover material is carefully placed so as not to puncture the membrane. The protective cover is usually at least 1 foot (0.3 meter) thick. Although developed as a canal lining, the asphalt membrane has been used extensively in all kinds of reservoirs (e.g., water supply, sewage lagoons, brine pits, fish ponds, and farm ponds).

15.2.2 Hydraulic Mix Design Requirements

There are certain basic requirements for HMA mixes to be used for waterproofing hydraulic structures. They must be workable while placing, and impermeable, flexible, and stable in service. To ensure impermeability (less than 1×10^{-7} centimeters per second), the void content of an asphalt concrete lining should not exceed 4 percent. A compaction level of 96 percent of the maximum specific gravity is required.

The lining will usually be placed on a sloped surface. Therefore, the mixture should be quite workable to allow efficient placement by both mechanical and hand methods. Usually, the desired compaction must be achieved using lighter compaction equipment. While operating on the slope, cables are used to winch the rollers up and down.

The lining should be flexible and able to withstand, without cracking, long-term loading stresses caused by settlement in the supporting subgrade or foundation. On the other hand, it must also be resilient under short-term loading, such as wave impact.

Increased asphalt content helps satisfy the foregoing design requirements. However, after a certain point, increased asphalt content results in decreased stability. Although this mix does not need the high stability usually associated with highway paving mixes, it should be stable enough to support its own weight on relatively steep slopes while hot. In addition, it should not creep after the lining has been compacted.

Impermeable Mix Composition

Mix design of impermeable asphalt mixes for hydraulic structures follows the same general principles established for pavement mixes. Use of the locally accepted mix design method is recommended. However, it is essential that the density of the compacted laboratory-prepared specimens be comparable to that which can be obtained in the field. It is suggested that not more than 35 blows of the compaction hammer on each end of the specimen be applied with the Marshall method. A range of 60–65 gyrations is used with Superpave compactor because the compaction effort on

a slope is considerably lower than that possible on roadway pavements. The design procedure consists of selecting the grade of asphalt cement, choosing a suitable aggregate gradation, preparing specimens at different asphalt contents, and conducting a density-void analysis for each sample.

In general, the Performance Grade (PG) of asphalt selected for these mixes should be one grade higher on the high-temperature side, when compared with a normal highway binder grade in that climate. For example, if a PG 64–16 is typically used, a PG 70–16 should be chosen for use in hydraulic linings. Only neat asphalts should be used for drinking water reservoirs. This relatively more viscous asphalt cement produces linings with added resistance to the destructive action of water, the growth of vegetation, and the extremes of weather. Linings made with this grade of asphalt cement are more stable on side slopes than linings made with the softer grades, yet they retain sufficient flexibility to conform to slight deformations of the subgrade.

The asphalt content will usually range from 6 to 9 percent, which is somewhat higher than the asphalt content of average roadway paving mixes. The asphalt content is limited, however, by the ability of the mix to resist flow down the slope at placement temperatures.

The aggregate used should be sound and have good adhesion characteristics in the presence of water. The maximum aggregate particle size should be limited to one-third to two-fifths of the finished thickness of the paving course to ensure uniform density throughout the layer. For mineral filler, crushed limestone dust is generally preferred. However, other inert mineral dusts having satisfactory performance records may be considered. Suggested mix compositions for asphalt concrete for impermeable linings are listed in columns A, B, and C of Table 15.8.

As a final check, a test or trial strip is recommended. It should be placed under the same conditions as will be experienced in the field. Careful observation of the behavior of the designed mix during the spreading and compacting operation might indicate slight adjustments in the job-mix formula that could not be detected in the laboratory.

Permeable Surface Mix Composition

Permeable asphalt mixes for hydraulic structures are characterized by the absence or reduced amount of fine aggregate or sand in the mix. As a consequence, the asphalt content is also reduced. The mixes have interconnected voids that permit passage of water. The void content of these mixes must be a minimum of 18 percent to assure easy passage of water through the mix.

A harder, or more viscous, grade of asphalt cement is desirable in these mixes to allow sufficient film thickness and to prevent drainage from the aggregate during construction. This choice of asphalt also provides additional cohesion in the mix between the aggregate particles.

The selected grade should be two PG grades higher on the high-temperature side compared with a normal highway binder grade in that climate. For instance, instead of using a PG 64–22, a PG 76–22 should be considered.

Table 15.8
Mix Compositions for Formed In-Place Asphalt Liners

Sieve Size		A	B	C	D	E
		Percent Passing				
1 in	(25.0 mm)			100	100	100
¾ in	(19.0 mm)		100	95–100	95–100	95–100
½ in	(12.5 mm)	100	95–100	-	85–95	-
⅜ in	(9.5 mm)	95–100	84–94	72–85	-	35–65
No. 4	(4.75 mm)	70–84	63–79	53–72	44–56	5–25
No. 8	(2.36 mm)	52–69	46–65	40–60	30–40	2–15
No. 16	(1.18 mm)	38–56	34–53	30–49	-	0–7
No. 30	(0.60 mm)	27–44	25–42	22–39	13–22	0–3
No. 50	(0.30 mm)	19–33	17–32	16–30	-	-
No. 100	(0.15 mm)	13–24	12–23	11–22	3–8	-
No. 200	(0.075 mm)	8–15	8–15	8–15	1–4	-
Asphalt cement, percent by weight of total mix		6.5–9.5	6.5–9.0	6.0–8.5	5.0–6.0	2.0–4.0
Minimum Recommended Compacted Depth		1 ½ in (37.5 mm)	2 in (50 mm)	3 in (75 mm)	3 in (75 mm)	3 in (75 mm)
Recommended Usage		Impermeable Surface	Impermeable Surface	Impermeable Surface	Permeable Surface	Drainage Layer

One method of determining the binder content is to mix up small batches with differing percentages of binder. Place the samples in Pyrex pie plates. Place the plates in an oven heated to mixing temperature. After 24 hours, observe the amount of drain off. The percentage of binder that shows only a slight drain off would be the proper binder content for that gradation of aggregate.

There are several analytical methods for determining the binder content of permeable mixes. Two of those methods include the following:

Federal Highway Administration Method

$$1.5 \times Kc + 4$$

where

Kc = Apparent Specific Gravity

Arizona Department of Transportation Method

$$\frac{(1.5 \times Kc + 3.5)2.65}{\text{Bulk Specific Gravity}}$$

where

Kc = Apparent Specific Gravity

Aggregate for permeable mixes, as with impermeable mixes, should be sound and have good adhesion characteristics in the presence of water. A high percentage of the coarse aggregate should be angular or crushed.

Suggested mix compositions for asphalt concrete for permeable linings are listed in column D of Table 15.8.

Sub-liner Drainage Layers

Porous drainage layers, with higher void content than a permeable lining, serve as drainage layers under an impermeable lining while at the same time contributing to the structural strength of the lining. In either case, the purpose is to provide free drainage to prevent hydrostatic pressures from building up in the embankment or within the lining itself. As a surface lining, water is allowed to flow to and from the embankment through the lining. As a drainage layer, the subsurface water should be collected and carried off by drains.

To be effective, porous drainage layers must have 20–22 percent void contents to assure proper drainage of water through the layer.

Asphalt binder selection and content is accomplished with the same procedures used for permeable mixes. The suggested mix composition for the porous layers is listed in column E of Table 15.8.

15.2.3 Construction Considerations

Impermeable Surface/Lining

Hydraulic linings of asphalt concrete should be prepared and placed in accordance with established construction practices. However, some special aspects of construction are mentioned here.

Asphalt concrete linings are placed on side slopes, on curved sections, in angles and corners, and on level surfaces. Conventional highway spreaders, modified pavers or spreaders, spreader boxes, special slip-form pavers, hand methods, or some combination of these methods may do this. In most instances, conventional highway spreaders are used. They are often modified as necessary for slope operations. When side-slope paving and floor paving are separate operations, experience has shown that best results are obtained when the slopes of the reservoir or channel are paved before paving the floor.

Conventional highway spreaders, (sometimes winched up the slope), usually perform spreading on side slopes; often they are modified to facilitate the work. In some cases, especially with large shallow reservoirs, asphalt mixes are spread with specially designed finishers operating on rails within a steel-frame structure up and down the slope. Also, side slopes of shallow reservoirs have been paved with pavers moving along the slope longitudinally. During the operation, the paver is held with cables attached to equipment operating on the crest. This method presents compaction problems, however, because rollers operating this way apply nonuniform rolling pressures and the drum tends to cut into the mix on the downslope side.

This problem can be eliminated, however, by using computer-controlled winches that maintain a constant tension in the anchor cables. Outriggers with wheels attached to the side of the rollers have also been used successfully. The outriggers can be hydraulically adjusted to keep the surface of the steel drum parallel and in uniform contact with the mat being laid.

When a spreader operates up and down the slope, a sloped ramp at the crest may be used to permit the paver to pave all the way to the top. Cable winches attached to the ramp framework raise and lower the paver on the slope. Upon completion of a paved strip with the paver on the ramp, the entire assembly is then moved over for the next paving pass.

When the paving operation proceeds down the slope, the ramp is not necessary. Recently, this has become the preferred method of paving the slope.

Several methods can be used to supply the paver with hot mix for slope paving. Probably the most common is to load the paver at the bottom of the slope with enough mix for one complete trip up the slope. Another is to load the paver at the top of the slope and then lower it to the bottom for the paving pass. When slope distances are too great for paving a strip without recharging, a front-end loader with a side-dump bucket has been used to supply the paver partway up the slope. Cranes have also been successfully used to supply the paver on the slope. Material transfer vehicles at the bottom of the slope, feeding conveyors equipped with belting material approved for handling hot materials, have also been used successfully.

Light steel-wheel rollers and vibratory steel-wheel rollers are often used for compacting the freshly spread mix on the slope. Power winches and cables are usually required for maneuvering the roller up and down the slope. The rolling is done independently of the paving, but it should begin as soon as possible without interfering with the paver. The initial rolling pass is usually made up the slope. Depending on the stability of the mix, the vibrator is not desired on the initial pass but on the following coverages.

When the lining is to be placed in two or more courses, the paving joints should be staggered or offset. Emphasis should be on a tight bond for impermeability.

A tack coat is usually required whenever an asphalt mix is placed over another paved surface. For bonding impermeable asphalt layers together, a tack coat of SS-1, SS-lh, CSS-1, or CSS-1h emulsified-asphalt diluted 1:1 with water and applied at a rate of 0.05–0.15 gallons per square yard (0.2–0.7 liter per square meter) over the surface to be paved will usually be adequate. Tack coats of emulsified asphalt should be allowed to cure fully before paving; otherwise moisture will be trapped between the two layers at construction.

Tack coats can be critical when used to bond two impermeable layers together. An excessive tack coat application combined with warm temperatures can cause a slippage plane between two layers on a sloped surface. On the other hand, unless the two layers are bonded together so that there are no voids or pores in the interface, seepage water from a crack or opened joint may get between the layers. If this happens, hydrostatic pressures can develop and eventually cause layer separation.

Permeable Surface/Lining

Essentially, the procedures for placing permeable surface lining mixes are the same as those for impermeable asphalt mixes. However, they are generally placed in one layer 3–6 inches (7.5–15 centimeters) thick, rather than in two or more courses.

There is no real effort to obtain density in open-graded drainage layer courses; the intent is only to firmly seat the aggregate particles. Excessive rolling can be harmful because of aggregate degradation or by the breaking of the asphalt bond between particles, particularly if the mix temperature is below 150°F (66°C). For this reason, generally, only a static steel-wheel roller is used.

Porous Drainage Layer

The same construction for permeable linings is used for porous drainage layers, with the expectation that the mixing temperature at the hot mix facility should not exceed 250°F (121°C). Higher temperatures may cause the asphalt to drain from the aggregate particles and settle in the bottom of the truck bed as it is hauled to the paving site. This problem is not usually encountered with mixes such as mix composition D.

Asphalt seals

An asphalt seal may be applied to an asphalt surface for a number of reasons. It may be designed to make the surface more watertight, or to protect it from abrasion by waves, water currents, or even by ice. It may also be used to protect the surface from mud curl or the curling of drying algae along the waterline, or to reduce the rate of oxidation of the exposed asphalt surface. These seals are generally of two types.

- **Sprayed asphalt sealers/asphalt cement.** (It should be noted that emulsified asphalt or cutback asphalt cannot be used as a seal for reservoirs that are used for drinking water, to avoid the residual emulsifying agent or cutter stock from contaminating the water.) The asphalt should be sprayed over the surface of an asphalt lining at the rate of 0.25 gallons per square yard (1 liter per square meter). This will provide a film coating as much as 0.04 inch (1 millimeter) thick. A continuous film coating will fill and seal any exposed pores and increase the water-tightness of the asphalt lining. It will also tend to fill and seal small cracks in the surface that may have been caused by improper rolling procedures in compacting the lining.

  The surface should be clean, dry, and free from loose material. A sloped surface usually necessitates hand spraying. This should be done in a back and forth sweeping motion to build up the film and to keep the asphalt from flowing down the slope.
- **Asphalt mastic seals.** Asphalt mastics applied to the surfaces of asphalt linings provide water proofing and extra surface protection from mechanical abuse. Asphalt mastics permit a heavier coating than sprayed asphalt applications, and well-designed mixtures can make the surface resistant to abrasion by waves, or scouring by waterborne sands.

Asphalt mastic mixtures for this purpose are essentially blends of mineral filler and asphalt cement. A small percentage (about 2 percent) of inert fibers is sometimes added to the mix to increase stability. The composition of the mastic for sealing can vary widely. However, it should usually fall within these ranges:

Mineral filler (typically lime)	50–70 percent
Asphalt Cement	30–50 percent

Such mixes require the use of a squeegee. Depending on the consistency of the mix, the surface thickness will be ⅜ inch (3 millimeters) or less. Fine sand may be added to give a gritty surface texture. When this is done, asphalt contents below 30 percent may be desirable.

These mixes usually are too liquid to prepare in a conventional HMA plant; special mixing equipment may be necessary.

The same grade of binder should be used in the mastic seal as was used in the mix for the liner, but lower viscosity grades can also be employed. Asphalt mastic seals are not easy to produce or to apply.

15.3 HMA Utilization in Railway Track Beds

15.3.1 Introduction

Since deregulation in 1980, the U.S. freight railroad industry has experienced unprecedented growth in traffic volumes, revenue ton-miles, and wheel loadings. This growth has necessitated the application of new technologies to provide stronger and longer-lasting track and support structures.

The railroad companies and the asphalt paving industry have developed designs and applications for utilizing a support layer of HMA to strengthen the track structure. HMA is being used in new mainline tracks, yards, and terminals. It is also being used as a solution to specific trackbed (maintenance) instability problems in existing trackbeds. These include the reconstruction of turnouts, railroad crossings, bridge approaches, hump tracks, tunnel floors and approaches, highway crossings, and loading facilities where conventional trackbed designs have not performed satisfactorily. The use of HMA in railway track beds results in an optimal track modulus, which significantly reduces the rail and rolling stock wear.

Two designs are used to incorporate HMA in the track structure as depicted in Figures 15.8 and 15.9. The most common design used on U.S. railroads is known as *HMA underlayment* in which the HMA layer is placed directly on select subgrade or old roadbed and the typical layer of ballast is subsequently placed between the HMA layer and the ties. This design requires little change from normal trackbed designs because the HMA layer merely replaces the typical granular subballast layer.

The HMA layer is typically 5–8 inches (125–200 millimeters) thick and 12–14 feet (3.7–4.3 meters) wide. It is similar to a paved lane of a highway, but is contained within the track structure to form a structural layer or hardpan between the ballast and subgrade or existing roadbed.

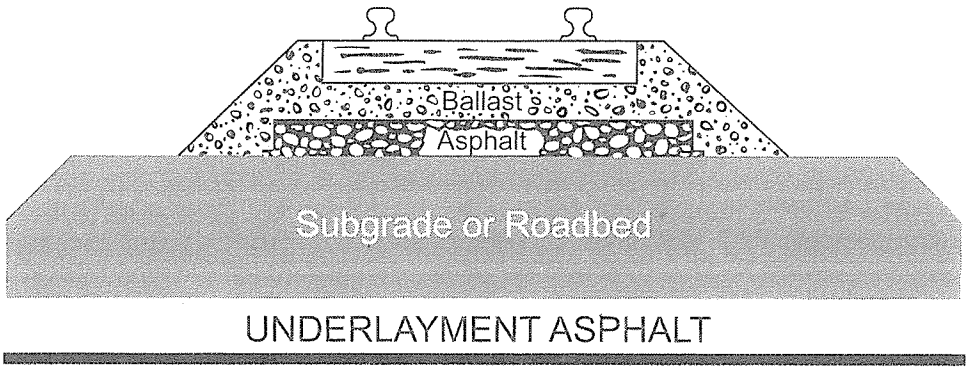


Figure 15.8 HMA Placed as an Underlayment

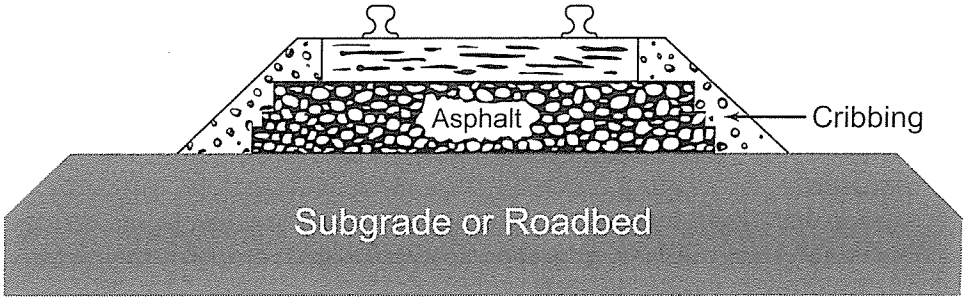


Figure 15.9 HMA Placed as an Overlayment

The HMA overlayment design involves placing the HMA layers in a similar manner except that no intervening ballast is used between the HMA layer and the ties (Figure 15.9). The ties are placed directly on the smooth, plain HMA surface. Cribbing aggregate is placed between and at the end of the ties to restrain track movement. This design has been used extensively with high-speed rail facilities. In the U.S., the HMA overlayment method has been limited to special applications, such as rail transfer yards.

15.3.2 Desirable Benefits and Applications of HMA Underlayment

An HMA underlayment has the following benefits:

- A strengthened track-support layer below the ballast to uniformly distribute reduced pressures to the roadbed (subgrade)
- A waterproofing layer and confinement to the underlying roadbed to provide consistent load-carrying capability for track structures even on roadbeds of marginal quality
- An impermeable layer to divert water to side ditches, essentially eliminating roadbed (subgrade) moisture fluctuations, to consistently maintain adequate underlying support
- A consistently high level of confinement for the ballast to provide high shear strength and uniform pressure distribution
- A resilient layer between the ballast and roadbed to reduce the likelihood of subgrade pumping without substantially increasing track stiffness
- An all-weather, uniformly stable surface to place the ballast and track superstructure

Track sites deemed most applicable for realizing the benefits of an HMA underlayment are those having heavy freight traffic and/or high-speed passenger traffic, where one or more of the following conditions exist or are expected:

- Difficulty in establishing and maintaining a sufficiently strong and stable roadbed (hardpan) to adequately support the ballast and track
- Difficulty in establishing and maintaining proper surface drainage to convey surface water away from the track structure
- Difficulty in preventing groundwater from weakening the track structure
- Abnormally high-impact stresses at railtrack crossings, joints, bridge and tunnel approaches, other special trackworks, or open track where track stiffness varies abruptly

Areas where these conditions exist are likely to show rapid track contamination, excessive wear of track components, and below-standard track geometric parameters. Maintenance costs become too expensive to continue safe line speed operations, so slow orders (that reduce operating efficiency) must be imposed.

15.3.3 Prevailing Design Practices

Recommended ideal HMA mix specifications, trackbed section designs, and application practices have steadily evolved over the years, reflecting refinements from initial recommendations. Slight variations from the recommended mix designs and construction techniques are typical and do not appear to adversely affect HMA trackbed performance.

The HMA mix, which is considered to impart the ideal properties to the track structure, is a low modulus (plastic) mix, having design air voids of 1–3 percent. This mix will easily compact to less than 5 percent air voids. This has been accomplished by specifying the local dense-graded highway base mix having a maximum aggregate size of 1.0–1.5 inches (25–37 millimeters). Ideally, the asphalt binder content can be increased by about 0.5 percent above optimum for highway applications because rutting and bleeding are not concerns in the insulated trackbed environment.

Long-term monitoring and testing of in-service trackbeds indicate that this low-voids, impermeable mix undergoes minimal oxidation from the effects of air and water. It is also isolated from extreme temperature fluctuation within the insulated trackbed environment. This isolation provides a layer with reasonably consistent stiffness in hot weather but slightly resilient in cold weather, thus eliminating any chance for the mix to rut and bleed in hot weather or crack in cold weather. A long life for the mix is thus ensured.

Tests on subgrade/roadbed samples, obtained directly from under the HMA layer, indicate that the in situ moisture contents are very close to optimum values for maximum density of the respective materials. For structural design analyses, it is reasonable to assume base support strength or bearing capacity values at optimum conditions for the material under the HMA layer.

The predominant HMA layer width is 12–14 feet (3.7–4.3 meters) for open track, but the width is wider under special trackwork, such as turnouts, to provide support under the longer ties.

HMA layer thickness varies depending on the quality of the roadbed (subgrade) support and traffic loadings. A layer 5 to 6 inches (125–150 millimeters) thick is normally used for average conditions. For unusually poor roadbed support conditions and high-impact areas, a minimum thickness of 8 inches (200 millimeters) is used. Ballast thickness usually is 6–12 inches (150–300 millimeters).

The HMA layer should extend several feet (meters) beyond the ends of special trackwork so that subsequent track surfacing operations and any impact from track stiffness changes will not infringe on the area. The roadbed should be reasonably well-compacted, well-drained, and capable of accommodating the hauling and spreading equipment without excessive rutting or deformation. A slight crown or side slope is desirable. The need for sub-surface drainage or roadbed support improvements, before placing the HMA, will depend on site-specific conditions.

The structural design of railway trackbeds containing HMA underlayment can be performed using KENTRACK, a finite element computer program available through the University of Kentucky Civil Engineering website. Asphalt Institute's Research Report RR 84-1 covers the use of this program and the design of railway systems utilizing HMA underlayments as an alternative to the traditional ballast systems. The primary governing factor is limiting the vertical compressive stresses, or permanent deformation, on the subgrade. For the HMA layer, the tensile strains at the bottom of the HMA layer control its service life. Damage analyses are conducted to predict the service life of the trackbed components for various combinations of traffic, tonnages, subgrade support, and component layer compositions and thicknesses.

15.3.4 Typical Installation Practices

The construction of *new* rail lines and extensions, classification yards, and intermodal terminals represent ideal conditions for HMA underlayment. These are usually large projects with exposed subgrades that are readily accessible to conventional asphalt paving equipment for placing the HMA layer before placing the ballast and track. Typical highway paving technology can be used. A typical paving view on the Burlington Northern-Santa Fe Railway's double-tracking of its transcontinental mainline in Southern Kansas is shown in Figure 15.10. This project involves approximately 200 miles (320 kilometers) of asphalt underlayment. The HMA is being placed in two lifts, with a total compacted thickness of 6 inches (150 millimeters).

New intermodal yards use HMA for all or portions of new track construction. A particular advantage is the waterproofing characteristics of the HMA layer and the positive drainage systems that can be utilized in the design of the loading area. Also, better construction accessibility reduces the construction time required.

The maintenance/rehabilitation of *existing* rail lines, using present technology, requires that the track be removed and the underlying material



Figure 15.10 Placing HMA Underlayment in Southern Kansas

excavated to the desired grade. The existing roadbed material should be reasonably compact and capable of providing adequate support. The HMA is hauled by dump trucks. For short track sections, the mix is commonly dumped and spread by motor grader to the desired configuration. Paving machines can be used for longer sections where maneuverability permits. Conventional compaction equipment is used. Placing the ballast and tracks completes the process. It is not necessary for the HMA to cool before placing the ballast. A typical view of placing an asphalt underlayment in an existing CSX Transportation rail crossing in northern Ohio is shown in Figure 15.11.

Also, mixes with reclaimed asphalt pavement and cold mixes have been used. Cold mixes can be prepared and stockpiled at the site for an extended time before placement in the trackbed.

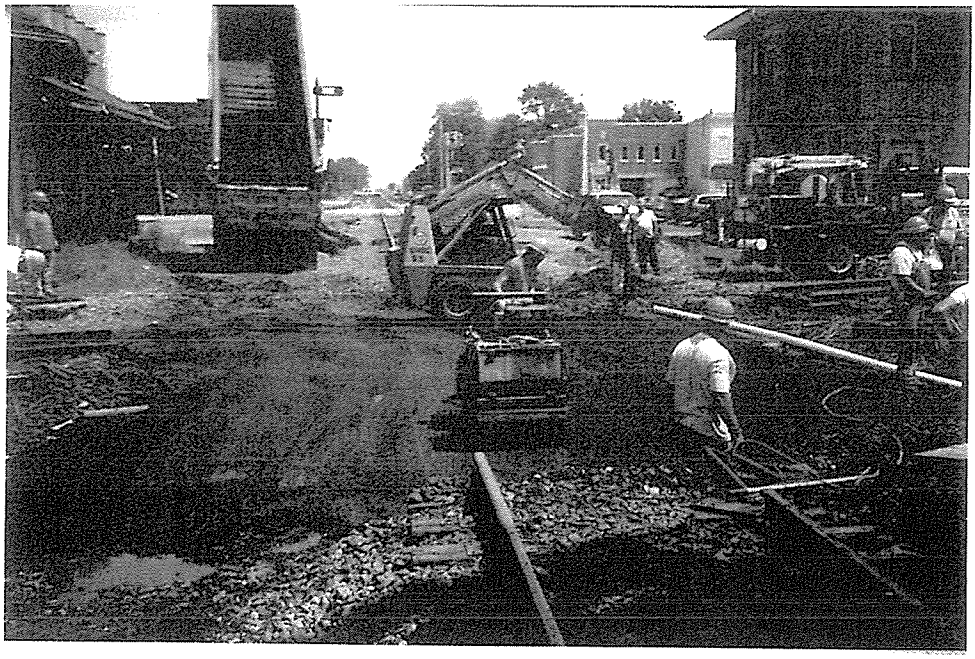


Figure 15.11 Placing Underlayment by Backdumping and Spreading

15.4 Athletic and Recreation Facilities

15.4.1 Types of Facilities

Almost every conceivable type of athletic or recreational facility can use asphalt in some way—from running tracks (see Figure 15.12) to walking paths to tennis courts (see Figure 15.13) to playgrounds to football fields. A wide selection of all-weather asphalt construction is economically available for these purposes. When asphalt is properly designed and constructed, maintenance costs are negligible. These surfaces are designed and constructed differently than a typical roadway or parking lot; when developing these types of facilities, a qualified consultant with experience in the particular type of facility (walking or running track, tennis court, etc.) should be engaged. Also, it is suggested that the contractor have experience with the construction of the type of facility being built.

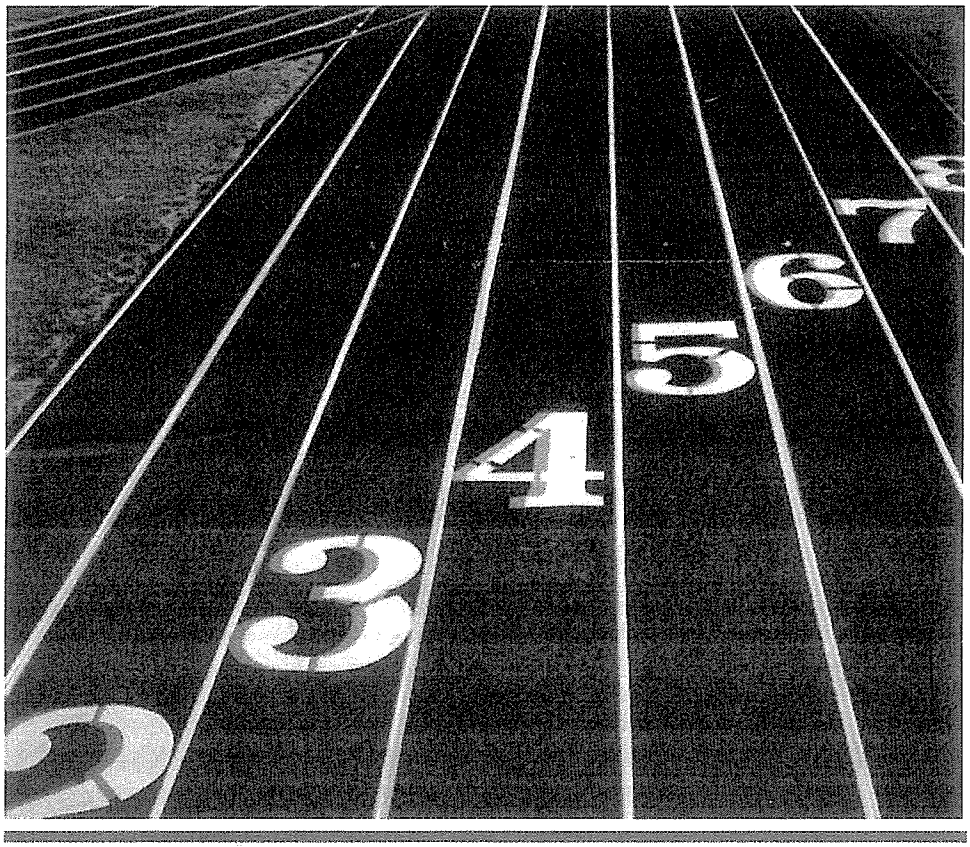


Figure 15.12 Asphalt Running Track

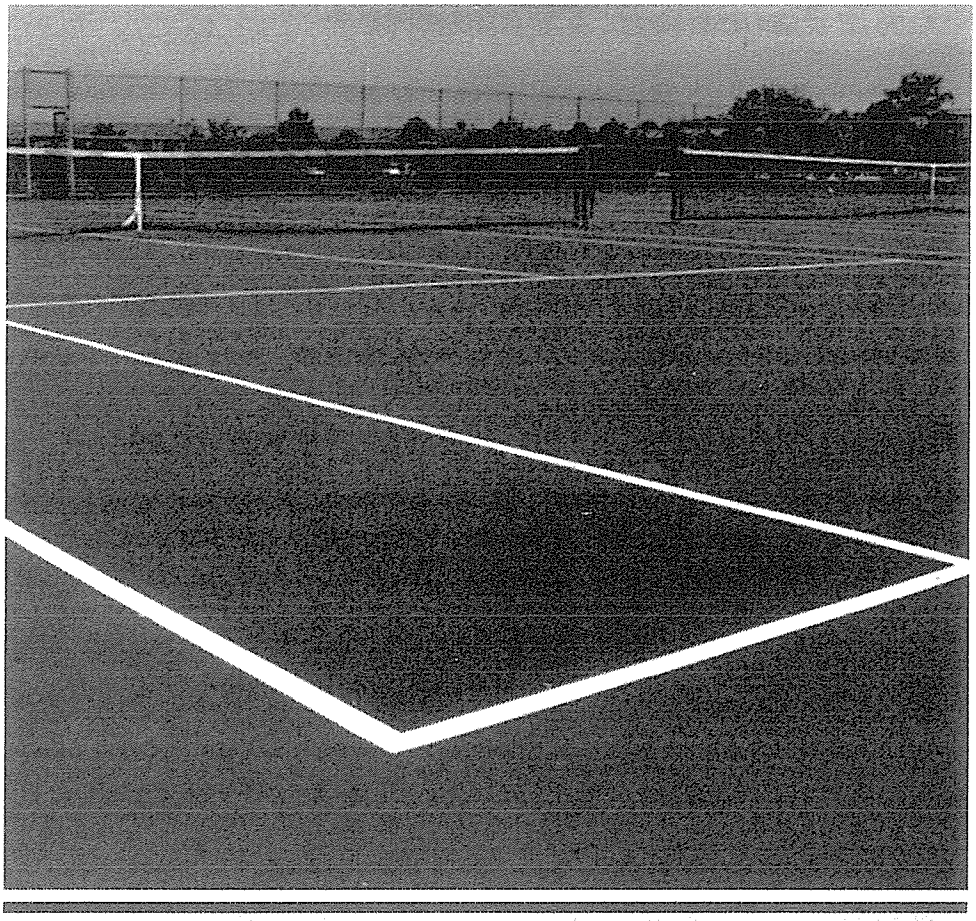


Figure 15.13 Asphalt Tennis Court

It is recommended that requirements of the U.S. Tennis Court and Track Builders Association be consulted.

15.4.2 Basic Considerations

A properly designed and constructed asphalt pavement can provide a long-lasting surface. The surface must be designed to meet the needs of the users and to accommodate local site conditions and materials. The existing terrain, drainage, and pavement uses of the facility must be addressed in the design phase. The basic construction must provide the following:

- A stable, well-compacted subgrade
- A free-draining, compacted, crushed-stone base
- A compacted HMA layer 3–4 inches (75–100 millimeters) thick that is placed in two lifts
- For running tracks and tennis courts, a special surface such as a liquid-applied acrylic for tennis courts or a rubberized surface for running tracks. For a walking path, no special surface will usually be required.